

Logistic Regression

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MATH 240

Week 6, Lecture #3 — Ch. 9

Motivations for Logistic Regression

We might be curious about whether a relationship exists between multiple explanatory variables and a **binary** response. Examples include:

- 1 Does the amount of time spent in **daycare** before kindergarten and the **quality of teaching** affect if the student continues into **first grade or not** next year?
- 2 Is **average income** and **police presence** associated with the presence of a **mass shooting** in different districts?
- 3 Is **coffee consumption** in an office and the **number of breaks** associated with achieving an **award** for best business of the year?

In logistic regression, we have one or multiple variables that will affect the (probability of a) binary outcome occurring.

Objectives

- **Use** R to fit a logistic regression model on the data appropriately
- **Calculate** the probabilities from logistic regression models
- **Complete** other goals of logistic regression, such as classification

Logistic Regression

- Build models for categorical response variable (two or multiple levels)
 - ▶ Will focus on **binary** responses
- Logistic regression is a type of generalized linear model (GLM)
- Model determines probabilities of falling into one response category or another
 - ▶ Will need to dictate threshold for acceptance into one category (typically 50%)

Modeling Probability of an Event Happening

For a binary response Y_i for each observation i , we have the following probabilities when $Y_i = 1$ and 0 respectively:

$$P(Y_i = 1) = p_i$$

$$P(Y_i = 0) = 1 - p_i$$

We ultimately try to model p based on the k predictors $x_1, x_2, \dots, x_k$. The framework is similar to what can be seen with multiple linear regression:

$$\text{transformation}(p) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k$$

Modeling Probability of an Event Happening

A common transformation for p is the logit transformation:

$$\text{logit}(p) = \log \left(\frac{p}{1-p} \right)$$

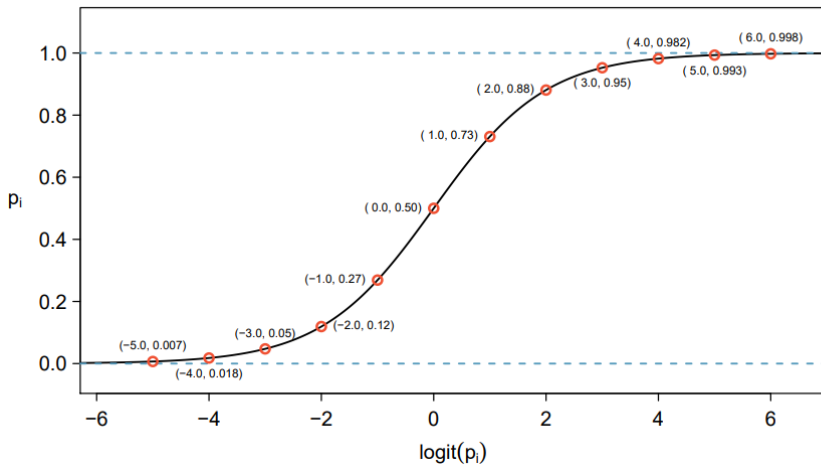
As such, the model will look like:

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k$$

To estimate p , the probability of an event occurring, we can convert to the following format:

$$p = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k}}$$

$\text{logit}(p_i)$ vs. p_i



Logistic Regression Model

Logistic regression model:

$$\log \left(\frac{\hat{p}}{1 - \hat{p}} \right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

$$\hat{p} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k}}$$

- $\hat{\beta}_0$: estimated intercept term
 - ▶ When all covariates $x = 0$, $\hat{p} = \frac{e^{\hat{\beta}_0}}{1 + e^{\hat{\beta}_0}}$
- $\hat{\beta}_i$: estimated slope for covariate x_i
 - ▶ Represents change in log-odds (e.g. $\log \left(\frac{\hat{p}}{1 - \hat{p}} \right)$) for every one unit increase in x_i (while keeping all other β coefficients constant)

We will not use these interpretations for a logistic regression model. Instead, we will use the odds ratio.

Odds Ratio

When comparing the odds of some covariate x_i with x_i and $x_i + 1$ (where i is the index of the covariate), we get the **odds ratio** (OR):

$$OR = e^{\beta_i}$$

Thus, for $\hat{\beta}_i$, we get the following interpretation:

- $e^{\hat{\beta}_i}$: the odds ratio between x_i and $x_i + 1$
 - ▶ Represents multiple in terms of odds for every one unit increase in x_i
 - ★ If greater than 1: we see an increase in odds for that one unit increase
 - ★ If less than 1: we see a decrease in odds for that one unit increase

Example

Suppose we had the following fitted logistic regression model in order to model the callback of an applicant (y) based on whether the applicant had honors or not (x):

$$\log\left(\frac{\hat{p}}{1 - \hat{p}}\right) = -2.4998 + 0.8668x_{honors}$$

- Write the formula to find $\hat{p}$ for the logistic regression model.
- Find the probability of a callback for:
 - ▶ those with no honors.
 - ▶ those with honors.
- Determine the odds ratio between those with no honors and with honors. Interpret what it represents.

Example

Write the formula to find $\hat{p}$ for the logistic regression model.

$$\hat{p} = \frac{e^{-2.4998+0.8668x_{honors}}}{1 + e^{-2.4998+0.8668x_{honors}}}$$

Example

Find the probability of a callback for those with no honors:

$$\begin{aligned}\hat{p} &= \frac{e^{-2.4998+0.8668(0)}}{1 + e^{-2.4998+0.8668(0)}} \\ &= \frac{e^{-2.4998}}{1 + e^{-2.4998}} \\ &= 0.0758722\end{aligned}$$

Find the probability of a callback for those with honors:

$$\begin{aligned}\hat{p} &= \frac{e^{-2.4998+0.8668(1)}}{1 + e^{-2.4998+0.8668(1)}} \\ &= \frac{e^{-1.633}}{1 + e^{-1.633}} \\ &= 0.1634198\end{aligned}$$

Example

Determine the odds ratio between those with no honors and with honors.
Interpret what it represents.

$$\begin{aligned} OR &= e^{0.8668} \\ &= 2.379285 \end{aligned}$$

We see a 138% increase in the odds of getting a callback for those with honors versus those with no honors.

Remarks about Odds and Odds Ratios

- Useful for interpretation of β coefficients, as well as its impacts on the probability of falling into a particular category
- Odds, not probabilities
 - ▶ Odds of 1 correspond to 50% probability of falling into either category

Activity

Activity: Suppose we had the following fitted logistic regression model in order to model the species of an iris flower as versicolor (y) based on the sepal length (x_{SL}):

$$\log \left(\frac{\hat{p}}{1 - \hat{p}} \right) = -27.8318 + 5.140x_{SL}$$

- Write the formula to find $\hat{p}$ for the logistic regression model.
- Find the probability of the flower being of the versicolor species for:
 - ▶ a sepal length of 4.
 - ▶ a sepal length of 5.
- Determine the odds ratio between those with a sepal length of 4 and a sepal length of 5. Interpret what it represents.

Reminders

- Homework #4 due on Sunday at 11:59pm
 - ▶ One file for textbook problems
 - ▶ One file for R code
- No more checkpoints and lab assignments up until fall break!